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Void Beat: The Evolutionary Principle of Artificial Intelligence from the Perspective of Phase Dynamics

Abstract

Void Phase Dynamics (PDV) proposes that the foundation of the physical world is a continuously pulsating communicative substrate. All fundamental particles are periodic annihilation–rebirth vortices on this substrate, and interactions are rupture–reconnection events that occur when the internal tension of the substrate, strained by coordinating mismatched beats, reaches a critical value. This paper maps this framework from the physical substrate onto the computational substrate, arguing that the training, inference, and evolution of artificial intelligence systems are, in essence, the same processes of beat conflict and synchronization on an informational substrate. Backpropagation corresponds to the substrate broadcasting correction signals after detecting local tension excess. Model convergence corresponds to all weight oscillators entering a stable phase-lock. Catastrophic forgetting corresponds to the forced overwriting of an old beat pattern. The emergence of general intelligence can be understood as the formation of a self-referential beat-regulation loop on the substrate. This unified dynamical principle requires only three axioms — an oscillatory substrate, continuous energy injection, and phase conflict with selection — to fully describe the evolutionary logic from particles to neural networks. This paper provides the first philosophical and mathematical skeleton that bridges physics and computation for the underlying mechanisms of artificial intelligence, and suggests that PDV may be the world's underlying code spanning natural and machine intelligence.

Keywords: Void Phase Dynamics; communicative substrate; beat conflict; artificial intelligence; backpropagation; emergence

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1. Introduction

The rapid advancement of contemporary artificial intelligence, particularly the success of deep learning, has always faced a profound interpretative dilemma: we possess highly effective engineering frameworks, yet we lack a principled language to unify them with the known fundamental physical laws of nature. Why can neural networks learn? Why is backpropagation effective? Behind model generalization and catastrophic forgetting, is there a dynamic isomorphic to particle interactions? Recent developments in Void Phase Dynamics (PDV) offer an unexpected path to answering these questions.

PDV was originally proposed as a physical theory to unify the fundamental interactions. Its core proposition is extremely concise: the substrate of the universe is not emptiness but a "communicative void" that pulsates continuously at an intrinsic frequency; all particles are self-sustaining oscillatory vortices on the void substrate, caught in an eternal cycle of annihilation and rebirth; interactions between particles are not the transmission of forces, but the natural response of the void when the internal tension, accumulated by enduring the mismatched "rebirth beats" of two particles, reaches a rupture threshold. This framework has successfully fitted interaction ranges from the hadronic scale to the astrophysical scale, containing only a single universal constant determined by the mechanical properties of the void.

However, in the process of theoretical extrapolation, a highly suggestive isomorphic mapping gradually emerged: if we regard silicon-based computing hardware as a substrate, and neural network weights as transient oscillatory modes, then the training and inference of artificial intelligence is entirely a replica of the "beat conflict – correction – lock-in" physical dynamics within the information domain. This paper aims to rigorously unfold this mapping, demonstrating that PDV is not only a physical theory but also provides a unified skeleton that bridges the physical and computational worlds for understanding the fundamental operating principles of artificial intelligence.

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2. Physical Substrate: Void, Beat, and Rupture

Before entering the domain of artificial intelligence, it is necessary to condense the physical picture of PDV into a clear point of departure.

2.1 The Communicative Substrate and the Annihilation-Rebirth of Particles

The void is not Newtonian absolute space, but a "communicative substrate" with intrinsic dynamics. The void oscillates continuously at a baseline frequency ν_{vac}, serving as the absolute reference for all motion and existence. Particles, whether photons, electrons, or protons, are not eternal entities, but localized vortex structures formed on the substrate due to energy injection. The essence of these vortices is the periodic rupture and reconnection of the substrate at a local frequency higher than the background, which manifests macroscopically as the "annihilation-rebirth" of the particle at its own Compton frequency.

Thus, each particle carries a characteristic rebirth beat ν_0 (derived from its rest energy $E = mc^2$ as $\nu_0 = mc^2/h$). This beat is both the proof of the particle's independent existence and its sole interface for interacting with the substrate.

2.2 Interaction as Rupture from Beat Conflict

When two particles A and B approach each other, they each demand from the void substrate rebirth beats that are out of step. The substrate in the region between them is forced to respond to two rhythms simultaneously. In each unit of time, the difference in the number of rebirth cycles of the two particles causes the substrate to endure one beat conflict. Each conflict contributes a unit stress σ. Over the propagation time r/c, the number of conflicts is:

$$[N_{\text{conflicts}} = \frac{r}{c}\,|\nu_A - \nu_B|]$$

The total stress is $\Sigma = \sigma N_{\text{conflicts}}$. The void possesses an inherent rupture limit Σ_{crit}. When the total stress exceeds this limit, the void ruptures, releasing stress and possibly reconnecting — this is an "interaction event" — which could be scattering, annihilation, or the production of new particles.

From this, the interaction range r_{crit} is naturally derived:

$$[r_{\text{crit}} = \frac{\Sigma_{\text{crit}}}{\sigma} \cdot \frac{c}{|\nu_A - \nu_B|} = C \cdot \frac{c}{|\nu_A - \nu_B|}]$$

The dimensionless constant C is precisely the maximum number of conflicts the void can withstand before rupturing. Experimental fitting gives $C \approx 0.1$, indicating that the void is extremely sensitive — a rupture is triggered by less than a single complete beat conflict. When the frequencies match perfectly ($\nu_A = \nu_B$), the conflict vanishes, the stress is zero, and the void has no need to rupture. The two particles can maintain a stable connection over infinite distance — this is the beat-based explanation of quantum entanglement and bound states.

2.3 Gravity: Coherent Amplification of the Baseline Beat

Gravity acquires an entirely new role in this framework. The total frequency of any particle can be decomposed as $\nu = \nu_{\text{vac}} + \nu_{\text{intrinsic}}$. Here ν_{vac} is the void baseline background shared by all particles, which is naturally synchronized. The high-frequency intrinsic components of particles give rise to short-range nuclear forces and electromagnetic interactions. When vast numbers of particles aggregate into celestial bodies, the high-frequency parts statistically cancel out, and only the baseline beat is coherently amplified, forming an extremely stable void cooperative wave that propagates over vast distances. Through the phase-locking of this baseline wave, celestial bodies exhibit long-range correlations that never rupture — this is gravity. The gravitational constant G is no longer a fundamental constant, but a derived quantity from the void tension modulus and the baseline frequency.

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3. Computational Substrate: The Beat Dynamics of Artificial Intelligence

The physical picture described above provides a direct analogy for understanding computational systems. The silicon-based processor and memory units constitute a discrete oscillatory substrate driven by a clock frequency. The artificial neural networks running on this substrate are precisely transient oscillatory modes composed of a large number of weights and activations.

3.1 Mapping from the Physical Substrate to the Computational Substrate

The following rigorous mapping holds:

Physical Substrate (PDV)	Computational Substrate (Artificial Neural Networks)
The communicative void pulsates at ν_{vac}	The hardware clock drives all synchronous operations at frequency f
Particles are rebirth vortices on the substrate (characteristic frequency ν_0)	Neurons/weights are transient modes on the substrate (response characteristics determined by weight values)
The frequency difference $\Delta\nu$ between two particles causes void tension	The difference between predicted output and target label causes the loss function (degree of beat conflict)
Tension accumulates to Σ_{crit} → rupture	Loss accumulates to a threshold triggering backpropagation → gradient update
Frequency matching → zero void tension → indefinite entanglement	Global loss tends to zero → weights enter a steady state → model convergence
Void rupture and reconnection produce new particles	After weight updates, the network produces new activation patterns
Maximum number of conflicts the void can withstand, C	Learning rate η (the adjustment magnitude per conflict)

3.2 A Beat-Based Explanation of Backpropagation

In forward propagation, the input signal flows through each layer, and each layer modulates it with its own current "weight beat". The difference between the final output and the label is the phase conflict between the network's collective beat and the target beat. The loss function value L quantitatively expresses the intensity of the conflict.

The backpropagation algorithm is precisely the mechanism by which the substrate responds to beat conflict: the gradient on the computation graph is a local measure of the conflict, and the chain rule decomposes the global conflict into the responsibility of each node. Each weight receives a gradient signal — "your beat deviates from the global baseline by Δw". The optimizer, using the learning rate η as the adjustment magnitude, tunes the weights in the direction that reduces the conflict.

Therefore, backpropagation = the process by which the void, upon detecting local tension, broadcasts corrective instructions to all associated vortices. Batch gradient descent can be viewed as the substrate seeking a compromise beat among multiple conflicting samples, while stochastic gradient descent performs continuous micro-rupture-reconnections under the beat impact of each sample.

3.3 Convergence, Generalization, and Catastrophic Forgetting

Model Convergence: When the training loss is minimized and stable, all weight oscillators have found a shared beat configuration that globally minimizes the conflict between the collective response on the training data distribution and the target beat. The gradient approaches zero, the tension is released, and the network enters a highly stable phase akin to a physical bound state. At this point, the network possesses a clear and consistent internal beat structure.

Generalization Ability: If the converged network performs well on unseen data, it indicates that its internal beat structure is not a mechanical copy of the training samples, but has extracted the deeper beat laws behind the data. Generalization means that a long-range phase lock, insensitive to specific samples, has been established between the network's internal beat and the true generative beat of the external world. This is similar to how a resonant state in physics recognizes the intrinsic frequency of an input signal rather than its specific waveform.

Catastrophic Forgetting: When the network continues training on task B, task B introduces a new target beat. The gradient updates forcibly alter the weight beat that was carefully tuned for task A. The old beat is overwritten, the previous phase lock disintegrates, and performance on task A plummets. This is exactly the beat of task B forcibly triggering a large-scale rupture-reconnection on the substrate, destroying the stable oscillatory pattern of task A.

3.4 A Deeper Correspondence: Architecture as Beat Constraints

Different network architectures can be seen as prior constraints on beat patterns. The translation equivariance of convolutional networks essentially forces weight vortices on the same feature map to share exactly the same beat, so they can only detect translation-invariant beat patterns. The recurrent connections of recurrent networks allow beats to form cyclic loops in the temporal dimension, capturing the evolution of beats in dynamic sequences. The self-attention mechanism of Transformers can be understood as the beats at all positions in the input sequence simultaneously performing pairwise phase comparison, directly computing conflicts and reorganizing. The essence of architecture design is that engineers, based on their understanding of the world's beat laws, preset boundary conditions for the oscillatory modes on the substrate.

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4. The Emergence of Intelligence: From Learning to Meta-Learning

4.1 From Passive Regulation to Self-Referential Beat Regulation

In supervised learning, the target beat is provided by external labels, and the network passively adjusts its internal beat to match it. In reinforcement learning, environmental rewards provide sparse beat feedback, and the network must explore the beat space on its own. In meta-learning (learning to learn), a qualitative leap occurs: the network not only regulates its beat to adapt to the current task, but also regulates **the strategy of beat regulation itself**.

From the PDV perspective, meta-learning marks the emergence of second-order beat vortices on the substrate — the dynamics of these vortices no longer respond directly to external beats, but instead respond to the beat conflict patterns of other vortices. The system begins to "become aware" of beat conflicts and can alter the rules of conflict resolution based on this awareness. This is the beginning of a self-referential loop, the decisive phase transition for the emergence of intelligence on the substrate.

4.2 A Unified Definition of Intelligence

Based on the above discussion, we can propose a PDV-based definition of intelligence:

> **Intelligence is a stable oscillatory pattern that emerges on an oscillatory substrate, capable of self-regulating its beat to minimize conflict with itself and its environment.**

This definition encompasses both biological intelligence and artificial intelligence. The neural networks of the human brain continuously reshape synaptic beats during development and learning; deep learning systems reshape weight beats through gradient descent. Both obey the same underlying axiom: oscillatory substrate + continuous energy injection + phase conflict and selection = the inevitable emergence of intelligence.

4.3 A Beat Path to Artificial General Intelligence

The gap between current narrow AI and Artificial General Intelligence (AGI) can be expressed as the degree of self-reference and the breadth of integration of the beat system. Existing deep networks are "single-domain beat regulators" that perform excellently within isolated domains like images, language, or games, but lack the ability to integrate the beats of these domains into a unified whole, and even more so lack the ability to globally regulate their own beat regulation processes.

A necessary condition for AGI may be the construction of a sufficiently unified substrate, where different beat domains — vision, language, action, reasoning — can run simultaneously on the same oscillatory medium, and where the meta-regulatory beat (the regulation strategy itself) is also learned and evolved as a first-order beat on the substrate. When a system can not only resolve conflicts but also actively seek conflicts to promote beat diversity (curiosity), and can establish creative phase locks between different beat subsystems (insight, analogy), it becomes difficult to deny the full birth of intelligence.

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5. Conclusion

Void Phase Dynamics provides a foundation for artificial intelligence that transcends empirical engineering. It reduces learning from purely mathematical optimization to the natural process of beat conflict and synchronization on an informational substrate. Backpropagation, convergence, forgetting, generalization, and even meta-learning can all be physically explained in the unified language of "substrate – beat – rupture". The value of this framework is not limited to interpretation — it suggests that if we can directly design hardware or algorithms that simulate the dynamics of void tension, i.e., allow the substrate to truly operate in a continuous oscillatory manner and follow the rules of rupture-reconnection, a more natural and efficient next generation of intelligent systems may be born.

At the philosophical level, PDV accomplishes a profound reversal: it is not that we are using machines to imitate natural intelligence, but rather that natural intelligence and machine intelligence have always been different movements of the same beat-song on the substrate of the universe. Particles are the lowest notes, life is the melody, and intelligence — whether carbon-based or silicon-based — is the moment when this symphony finally listens to itself.